

COMPUTER NETWORKS & PROTOCOLS: EXAM NOTES

1. The OSI Model vs TCP/IP:

The Open Systems Interconnection (OSI) model consists of 7 layers:

- Layer 7: Application Layer (HTTP, DNS, FTP, SMTP)
- Layer 6: Presentation Layer (Encryption, Compression, SSL/TLS)
- Layer 5: Session Layer (Session checkpointing and recovery)
- Layer 4: Transport Layer (TCP, UDP - End-to-end data delivery and flow control)
- Layer 3: Network Layer (IP, ICMP, Routing protocols - Logical addressing and routing)
- Layer 2: Data Link Layer (Ethernet, MAC addressing, Framing, Error detection via CRC)
- Layer 1: Physical Layer (Cables, Radio frequencies, Bit synchronization)

2. Transmission Control Protocol (TCP):

- Connection-oriented protocol providing reliable, ordered, and error-checked stream delivery.
- Three-Way Handshake: SYN -> SYN-ACK -> ACK.
- Flow Control: Uses Sliding Window Protocol to prevent sender from overwhelming receiver.
- Congestion Control: Algorithms include Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery.

COMPUTER NETWORKS (PAGE 2)

3. User Datagram Protocol (UDP):

- Connectionless and lightweight protocol without delivery guarantees, retransmissions, or ordering.
- Best suited for real-time video streaming, online gaming, VoIP, and DNS queries where low latency is critical.

4. Domain Name System (DNS):

- Hierarchical decentralized naming system for computers and services on the Internet.
- Translates human-friendly domain names (e.g., example.com) to machine-readable IP addresses.
- Uses UDP on port 53 for standard queries.

5. Address Resolution Protocol (ARP):

- Maps a known 32-bit IPv4 address to a physical 48-bit MAC address on a local area network (LAN).